

PRIVACY-PRESERVING HUMAN PRESENCE & MOTION DETECTION USING WI-FI CSI

Camera-free occupancy and activity sensing with ESP32-S3 and machine learning

OBJECTIVES

- ✓ Detect whether a room is occupied using only Wi-Fi CSI: no camera, microphone or wearable
- ✓ Tell a still person (sitting, standing) from a moving one (walking)
- ✓ Run live on low-cost ESP32-S3 boards, with results on a phone

KEY FEATURES

- ✓ **1 transmitter + 2 receivers:** ESP32-S3 boards, 50 CSI packets/s on Wi-Fi channel 6
- ✓ **Clean signal:** 107 subcarriers per packet, aligned on a 25 Hz grid, cut into 2-s windows
- ✓ **Two-stage Random Forest:** presence (31 features), then motion (44 features)
- ✓ **Self-calibrating:** 12-s empty-room calibration at every start-up
- ✓ **Phone dashboard:** START/STOP and live EMPTY / STATIC / MOVING

RESULTS

Tested once on a person never seen in training (subject S03)

Phase 1: EMPTY vs PERSON

99.8%
accuracy

100%
person recall

9 / 9
sessions correct

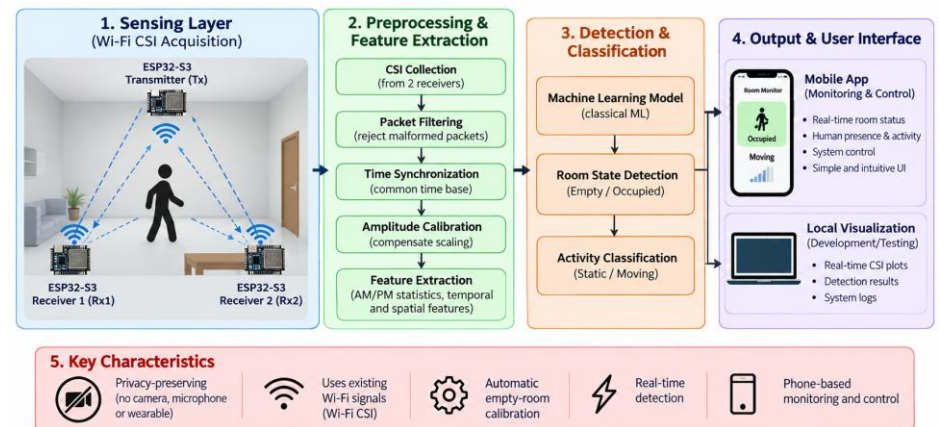
Phase 2: EMPTY / STATIC / MOVING

93.1%
window accuracy

99.2%
EMPTY recall

10 / 10
sessions correct

Dataset: 39 sessions · 78 CSV files · 3 subjects · 227,505 CSI packets



System Architecture



Hardware Implementation



Live Phone Dashboard

Conclusion: three low-cost ESP32-S3 boards detect presence with 99.8% accuracy and tell still from moving people with 93.1% window accuracy on an unseen person, without any camera.



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GitHub



Demo Video



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